TEXORA · LECTURE SHEET

Twill Weaves and Their Derivatives

In a twill, each pick shifts the interlacing pattern by one end, so the binding points march diagonally across the cloth and form the unmistakable twill line. From that one rule come the warp-faced, weft-faced and balanced twills of denim and gabardine — and, by pointing, breaking and combining, the pointed, broken, combined, steep, reclined and diamond derivatives.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 3 · LEVEL 3 TERM 2

1. The twill rule and the twill line

A twill is defined by a **run**: a fixed sequence of ends up and down, repeated along each pick, and **stepped by one end on every successive pick**. The weave is written as a fraction, e.g. **2/2** (two up, two down), **2/1**, **3/1**, **1/3**: numerator = ends raised, denominator = ends lowered, and

repeat of a regular twill = numerator + denominator (ends = picks)

The stepped binding points join up into diagonal **twill lines**. If the line climbs to the **right** the twill is a **Z-twill** (right-hand); to the **left**, an **S-twill** (left-hand), following the direction of the central stroke of the letter. When ends and picks per cm are equal and the step is one end per pick, the line runs at **45°**; unequal densities tilt it (Section 7). Which system dominates the face follows the run: **3/1 is warp-faced** (denim), **1/3 is weft-faced**, **2/2 is balanced** (equal warp and weft show). Twills interlace less often than plain weave, so for the same yarns they weave **closer setts, softer and heavier cloth, with longer floats** — and correspondingly more seam slippage and less stability.

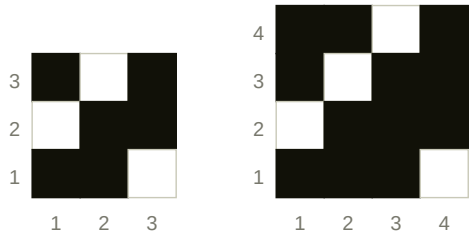


Figure 1 — Left: 2/1 twill (repeat 3 × 3), warp-faced. Right: 3/1 twill (repeat 4 × 4), strongly warp-faced — the denim construction. In both, each pick repeats the run stepped one end to the right (Z-twill).

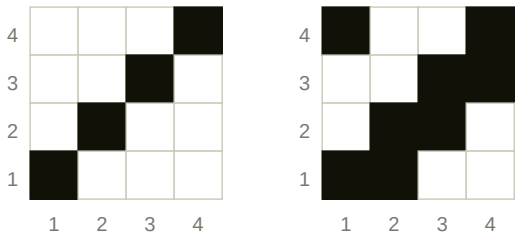


Figure 2 — Left: 1/3 twill (repeat 4 × 4), weft-faced; the warp binds only once per repeat, so the weft forms the face. Right: the balanced 2/2 twill (repeat 4 × 4) — the most used twill of all, the base of gabardine, serge and flannel.

2. Drafting and lifting for twills

A regular twill on a **straight draft** needs as many shafts as the repeat has ends (3 shafts for 2/1, 4 for 2/2 and 3/1), and its peg plan is the design itself read shaft by shaft: shaft *k* is lifted exactly where end *k* is marked. This is why twills are the natural teaching example of Topic 1's rule — with a straight draft, design and peg plan are the same pattern. Reversing the stepping direction in the peg plan turns a Z-twill design into an S-twill fabric without redrawing the draft.

3. Pointed (zigzag) twill

Keep the twill lifting, but draft the ends **pointed**: 1, 2, 3, 4, 3, 2, then repeat. The twill line climbs while the draft advances, then descends as the draft returns — a horizontal zigzag (chevron) band across the fabric.

warp repeat of pointed twill = 2 × (base repeat) − 2 ends; picks repeat unchanged

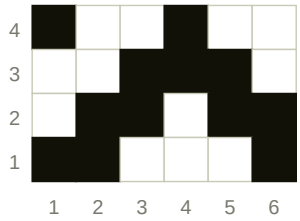


Figure 3 — Pointed twill from the 2/2 base (draft 1-2-3-4-3-2): warp repeat 6 ends = 2 × 4 − 2, picks 4. The diagonal advances for four ends and retreats for two steps, meeting its mirror at the point.

Herringbone is the close relative in which, instead of the draft turning, the *twill direction itself* is reversed in blocks of ends (the line changes from Z to S at a sharp break): the surface shows the broken, feather-like stripes of herringbone suiting. The distinction to state in the examination: pointed twill reverses the **draft**; herringbone reverses the **twill line direction** — visually similar zigzags, different construction records.

4. Broken twill

At a chosen pick the regular step is interrupted — the sequence jumps (e.g. after half the repeat, the order of lifting reverses or skips), so the continuous diagonal is **broken** and cannot form a visible line. The classic 2/2 broken twill uses the lifting order 1, 2, 4, 3 over the four picks instead of 1, 2, 3, 4.

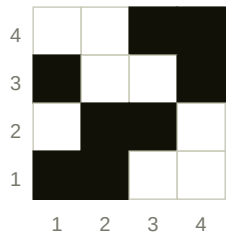


Figure 4 — Broken twill on the 2/2 base (lifting order 1-2-4-3). Compare with Figure 2 (right): the same marks per pick, re-sequenced so the diagonal is destroyed. The cloth keeps twill's sett and handle but shows no line — used for workwear and shirting where a line is unwanted.

5. Combined twills

Two different twills are woven in alternating bands — by ends (warp-way stripes of 2/1 and 1/2, say) or by picks (weft-way bands) — or are **merged into one repeat**: the runs of two twills interleaved end by end to make a new, often steep-looking pattern (the principle behind many fancy suiting twills and **satin-back** type combinations). The construction rule is bookkeeping: the combined repeat is the LCM of the parts in the direction of combination, and every component must be drawn out fully before the merge is trusted — exactly the discipline of the stripe design in Topic 5.

6. Diamond twill

Point the draft *and* point the lifting: the zigzags in the two directions cross and enclose lozenge (diamond) figures — the basis of diamond-pattern suitings, quilting designs and the ground of many honeycombs (Topic 5).

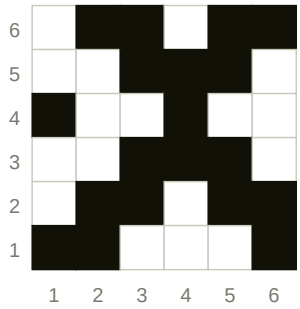


Figure 5 — Diamond twill from the 2/2 base: pointed draft 1-2-3-4-3-2 with pointed lifting in the same order — repeat 6 × 6. Where the two pointings meet, the twill steps reverse in both directions and the diagonal encloses diamonds; note the float of 3 that forms at the reversal points (longer than the base twill's 2).

7. Steep and reclined twills — the angle is a variable

The 45° line is a consequence, not a law. Two constructions change it mechanically. In a **steep twill** the step is made every *second* end (or the draft skips), so the line climbs faster than it advances: angle greater than 45°. In a **reclined twill** the same lifting is held for *two successive picks* before stepping, so the line advances more slowly: angle less than 45°. Figure 6 shows the reclined construction on the 2/2 base (repeat 4 ends × 8 picks). Density change has the same effect physically — doubling the picks per cm at fixed ends per cm halves the tangent of the angle — which is why a design drawn square on paper rarely lies at exactly 45° in cloth.

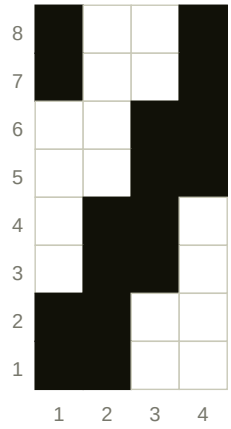


Figure 6 — Reclined twill, 2/2 base: each lifting is repeated for two picks before the step, doubling the pick repeat to 8 and flattening the twill line. The steep twill is the mirror construction (step of two ends per pick).

WORKED CHECK — TWILL ANGLE FROM DENSITIES

For a regular twill stepping one end per pick, each step moves one end-spacing sideways (1 ÷ ends per cm) and one pick-spacing upward (1 ÷ picks per cm), so, measuring the angle from the warp direction, tan(angle) = (ends per cm) ÷ (picks per cm). A 2/2 twill woven at 28 ends/cm and 24 picks/cm: tan = 28/24 = 1.167 → angle ≈ **49.4°** — a steepened line in cloth although the paper design is square. At equal densities the angle is exactly 45°; at 24 ends and 28 picks it would fall to ≈ 40.6°, reclined (denser picks pack the steps closer vertically, so the line climbs more slowly). This is the calculation behind "the same design, a different fabric."

Glossary

Twill line — the diagonal formed by stepped binding points. **Z/S twill** — line climbing right / left. **Run** — the up/down sequence of one repeat (e.g. 2/2). **Warp-faced / weft-faced / balanced** — numerator greater / less than / equal to denominator. **Pointed twill** — pointed draft, straight lifting. **Broken twill** — step interrupted so no line forms. **Diamond twill** — pointed draft and pointed lifting. **Steep / reclined** — line angle above / below 45°.

Key points

• Twill = a run stepped one end per pick; repeat = up + down; the fraction names the facing.

• Pointed: warp repeat 2R − 2. Broken: re-sequenced lifting kills the line. Diamond: both records pointed; floats lengthen to 3 at the turns.

• Angle follows the step construction and the densities: tan(angle) = EPI ÷ PPI for the one-step twill (angle from the warp direction).

Practice

1. Write the run and repeat of a twill that is weft-faced with the weft floating over 3 ends, binding on the fourth. Draw its design.
2. A pointed twill is built on a 3/1 base. State the warp and pick repeats.
3. The lifting order of a 2/2 twill is changed to 1, 2, 3, 4, 3, 2, 1, 2… over picks (pointed lifting) on a straight draft. Name the resulting family and state the pick repeat.
4. A 2/1 twill at 30 ends/cm and 30 picks/cm is re-woven at 30 ends/cm and 20 picks/cm. What happens to the twill angle and to the fabric's weft cover?

Answer key

1. Run 1/3 (one end up, three down), repeat 4 × 4; the design is the transpose-step pattern of Figure 2 (left): one mark per row, stepping one end per pick.
2. Base repeat 4 → warp repeat 2 × 4 − 2 = **6 ends**; picks remain **4**.
3. The pointed (zigzag) construction produced from the lifting side — the transpose arrangement of the pointed twill: the zigzag runs in the pick direction instead of the end direction. Pick repeat = 2 × 4 − 2 = **6**: the lifting order 1-2-3-4-3-2 then repeats from 1, exactly as the given sequence continues. (Point both draft and lifting and the two zigzags cross into the diamond of Figure 5.)
4. tan(angle) = ends ÷ picks rises from 30/30 = 1 to 30/20 = 1.5, so the angle steepens from 45° to ≈ 56.3° (each pick-spacing is now larger, so every step climbs further); with fewer picks/cm the weft cover also drops and the cloth becomes more warp-dominant and lighter per unit area.